sysMonitorData

Service Description

Service ID: *"sysMonitorData"*

Abstract
This document describes the `sysMonitorData` service, version 1.0.0, produced by the SysMonitor support system of Arrowhead 5.2.2.

Contents

1 Overview **3**
1.1 Significant Prior Art . . . 3
1.2 How This Service Is Meant to Be Used . . . 3
1.3 Important Delimitations and Dependencies . . . 3

2 Service Interface **4**
2.1 operation subscribe . . . 4

3 Information Model **5**
3.1 struct StoredObservation . . . 5
3.2 Primitives . . . 5

4 References **6**

5 Revision History **7**
5.1 Amendments . . . 7
5.2 Quality Assurance . . . 7

1 Overview

This document describes the `sysMonitorData` service, which provides the stream of all received observations, republished on the local MQTT broker. It is produced by the SysMonitor system [1] in service version 1.0.0.

The rest of this document is organized as follows. In Section 2, we describe the abstract message operations provided by the service. In Section 3, we end the document by presenting the data types used by the mentioned operations. The concrete interface is specified in the MQTT/TLS/JSON IDD [2].

1.1 Significant Prior Art

New in this workspace (SysMonitor-Data-MQTT-IDD-5.2.2).

1.2 How This Service Is Meant to Be Used

Consumers such as AIGym orchestrate `sysMonitorData` and subscribe to `arrowhead/<Cloud>/sysmonitor/data/<observationType>/<providerSystem>`.

1.3 Important Delimitations and Dependencies

QoS 1 from a transactional outbox; duplicates are possible after a crash between PUBACK and outbox deletion. Broker ACLs decide who may subscribe.

2 Service Interface

This section describes the interfaces to the `sysMonitorData` service. In particular, each subsection names an abstract operation, an input type and an output type, in that order. The input type is named inside parentheses, while the output type is preceded by a colon. Input and output types are only denoted when accepted or returned, respectively, by the interface in question.

All abstract data types named in this section are defined in Section 3.

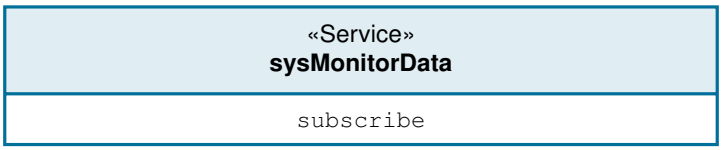


Figure 1: SysML-style block of the `sysMonitorData` service and its operations.

The following interface operations are available.

2.1 operation subscribe () : StoredObservation

3 Information Model

Here, all data objects that can be part of the `sysMonitorData` service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is used to denote a collection of named fields, each with its own data type. As a complement to the explicitly defined types in this section, there is also a list of implicit primitive types in Section 3.2, which are used to represent things like hashes and identifiers.

3.1 struct StoredObservation

Field	Type	Description
schemaVersion	String	`1.0`.
observationId	Identifier	Identifier (publisher's, or a SHA-256 of routing key and content).
providerSystem	String	Publishing system.
serviceInstance	String	Service instance.
observationType	String	Observation type.
sourceTime	DateTime	Publisher time, UTC.
receivedTime	DateTime	Time SysMonitor received it.
publishedTime	DateTime	Time it was forwarded on `sysMonitorData`.
value	Map<String, Object>	Observed value.
unit	String	Unit.
quality	String	Quality.
pipeline	Map<String, Object>	Intake details (routing key, exchange).

3.2 Primitives

Types and structures mentioned throughout this document that are assumed to be available to implementations of this service. The concrete interpretations of each of these types and structures must be provided by any IDD document claiming to implement this service.

Type	Description
DateTime	Pinpoints a specific moment in time (UTC).
Identifier	A string that identifies a record, e.g. a UUID or a composite identifier.
Map	A set of key-value pairs.
Object	Any structured value.
String	A chain of characters.

4 References

[1] J. Delsing, “SysMonitor – System Description,” Eclipse Arrowhead, SysD 5.2.2, 2026, ah5-SysMonitor-java-spring/documentation/v5.2.2/SysMonitor-v5.2.2-SysD.pdf.

[2] ——, “sysMonitorData MQTT/TLS/JSON – Interface Design Description (SysMonitor),” Eclipse Arrowhead, IDD 5.2.2, 2026, sysMonitor-v5.2.2-sysMonitorData-MQTT-IDD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	